



STA 201: Elements of Statistics and Probability

Lecture 1 & 2

Suborna Sultana

Adjunct Lecturer

Department of Mathematics and Natural Sciences

BRAC University

What is “Statistics”?

- It is difficult to define statistics in a few words; since its dimension, scope, function, use and importance are constantly changing over time. No formal definition thus has emerged so far and no definition is perhaps beyond controversy.
- According to [Sir R. A. Fisher](#) (1947), the science of statistics is essentially a branch of applied mathematics and may be regarded as mathematics, applied to observational data.
- Statistics is the science concerned with developing and studying methods for [collecting, analyzing, interpreting and presenting data enabling drawing valid conclusion on these](#).

The main objectives of statistics:

- Collecting the data
- Summarizing the data
- Estimating parameters from the data
- Testing the hypothesis
- Interpreting the results
- Drawing valid conclusion
- Forecasting the future values

Scope and applications of statistics:

The scope and uses of statistics are so wide and universal that they can't be enumerated instantly in a few words. Statistics has now been recognized as a separate discipline of human knowledge in its own right.

• Economics • Social Science • Business • Politics • Psychology • Engineering	• Medicine • Genetics • Environmental Science • Geology • Geography • Astronomy
---	---

Statistics has its extensive application in the following fields:

1. Surveys:

- Determine which political candidate is more popular.
- Discover what foods teenagers prefer for breakfast.
- Estimate the number of children living in a given school area.

2. Government Operation:

Government often conducts experiments to aid in the development of public policy and social programs. Such experiments include:

- Fluctuations in the economy
- Population trends
- Opinion polls.

3. Scientific research:

Statistical methods are used to enhance the validity of inference in all the fields of science, medical science etc. Such as:

- Radio carbon dating to estimate the risk of earthquakes.
- Clinical trials to investigate the effectiveness of new treatments.
- Field experiments to evaluate the irrigation methods.
- Measurements of water quality

4. Business and Industry:

Statisticians using statistical tools quantify the unknowns in order to optimize resources. They:

- Predict the demand for product and services.
- Check the quality of items manufactured in a facility.
- Manage investment portfolios.

Statistics in the Business World:

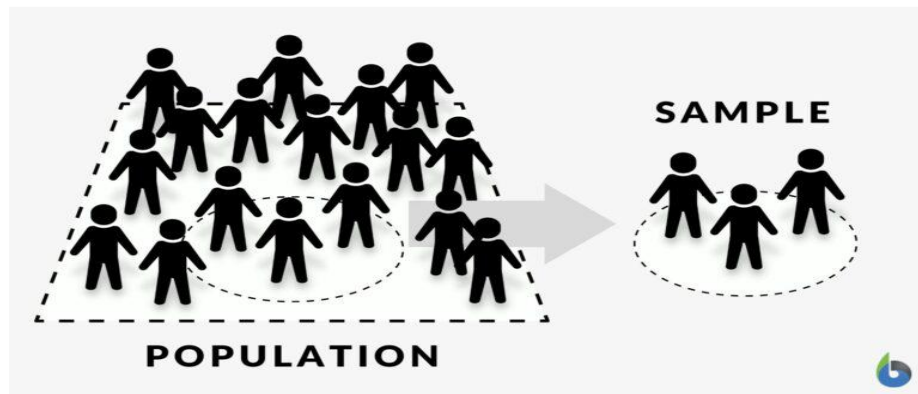
In the business world, statistics has four important applications:

- To summarize business data
- To draw conclusions from that data
- To make reliable forecasts about business activities
- To improve business process.

Statistics in Engineering:

- Electrical and Electronic Engineering/ Electronic and Communication Engineering:
 - Signal processing
 - Information theory
 - Reliability
 - Image processing
 - Power grid simulation and power grid analysis
 - Telecom engineering
- Computer Science/ Computer Science & Engineering:
 - Theoretical Computer Science
 - Machine Learning
 - Computer Vision
 - AI, Robotics
 - Natural language processing
 - Security

Some Basic Definitions of Statistics



Population:

- **Population is the collection of all individuals or items under consideration in a statistical study.**
- **Example:** If we want to study the average weight of the student of BRAC University then the set that consists of all the weights of the student of BRAC University will be the population in this case.

Parameter:

A parameter is a numerical measure that describes a characteristic of a population.

Sample:

- **A small and representative part of population is known as sample.**
- **Example:** If we want to study the average weight of the student of BRAC University but collected data from only 1st semester students then the set that consists of weights of 1st semester will be the sample in this case.

Statistic:

A statistic is a numerical measure that describes a characteristic of sample.

Exercise:

- If we want to determine the average height of the 2nd semester students of the BRAC University then what is the population and sample?
- A researcher wants to estimate the average weight of newborns in Bangladesh in the last five years. He takes a sample of 500 newborns from the records of two famous maternity hospital in Dhaka and obtains an average of 3.27 kilograms. Identify the population, sample, parameter and statistic for the above study.

Types of Statistics:

1. **Descriptive statistics** is the branch of statistics that involves organizing, summarizing, displaying, and describing data.
Descriptive statistics includes the construction of graphs, charts, and tables and the calculation of various descriptive measures such as averages, measures of variation, and percentiles.
2. **Inferential statistics** is the branch of statistics that involves drawing conclusions about a population based on information contained in a sample taken from that population and measuring the reliability of the conclusions.

Variable:

- **A characteristic that varies from person to person, place to place or time to time is called a variable.**
- Example: Height, weight, age, SSC and HSC marks, family size, etc.

Types of Variables: There are two types of variables -

1. **Qualitative variable (also known as categorical variable or attribute)**
 - A qualitative variable is one for which numerical measurement is not possible.
 - In other word when the characteristic being studied is non-numeric, it is called a qualitative variable or an attribute.
 - Example: **Hair color** (brown, black, white etc.), **religion** (Muslim, Hindu, etc.), **gender** (male, female), **home district** (Dhaka, Rajshahi, Bogra etc.), **occupational status** (employed, unemployed, self-employed, others) etc.
2. **Quantitative variable (also known as numerical variable)**
 - A quantitative variable is one for which the resulting observations have numeric value and thus possesses a natural ordering.
 - Quantitative variables are measured on a numeric or quantitative scale.
 - Example: **Height, weight, age, districts population size, customer's shoe size, vehicle's speed on a highway, amount of daily sell of a shop, parents income etc.**

Types of Quantitative Variables:

1. **Discrete variable:**
 - When a variable can assume only isolated values within a given range is called discrete variable.
 - Example: **Number of children in a family, number of road accident in a year, number of phone call received in a phone booth etc.**

2. Continuous variable:

- A variable is said to be a continuous variable if it can theoretically assume any value within a given range or ranges.
- Example: **height of a person – since it can take any value between 5.2 feet and 5.8 feet.**

Exercise:

1. Classify each variable as qualitative or quantitative:

- i. Marital status of nurses in a hospital
- ii. Time it takes to run a marathon
- iii. Weights of lobsters in a tank in a restaurant
- iv. Colors of automobiles in a shopping center parking lot
- v. Ages of people living in a personal care home

2. Classify each variable as discrete or continuous:

- i. Number of pizzas sold by Pizza Express each day
- ii. Lifetimes (in hours) of 15 iPod batteries
- iii. Weights of the backpacks of first graders on a school bus
- iv. Number of students each day who make appointments with a mathematics tutor at a local college
- v. Blood pressures of runners in a marathon

Data:

- Information obtained by observing values of a variable is called data.
- In a statistical analysis the first work is to collect data.

Sources of data: There are generally two sources of data-

Primary data:

- Primary data is obtained from an investigation conducted for the first time. Thus, the data collected for the first time by the investigator as original data are known as primary data.
- Example: The national census data collected by the government.

Secondary data:

- Secondary data is the data which is obtained from a prior investigation or which is already available.
- Example: Bangladesh Demographic and Health Survey (BDHS) dataset collected by the government are primary data but they become secondary data for those who use them.

Scales of Measurement

- Measurement is a process of assigning numbers to some **characteristics** or **variables** or **events** according to scientific rules.
- Variables in a study may be of different nature and they may represent some attributes, characteristics or key factors of interest.
- These variables can be measured under four levels or scales of measurement:

Types of Variable	Measurement Scale
Qualitative	1. Nominal 2. Ordinal
Quantitative	3. Interval 4. Ratio

Nominal Scale:

- A nominal scale simply names categories that values for the variable can fall within. Nominal = name. We can name the categories, but they have no natural order.
- The nominal scale is the lowest level of measurement because we can only group the observations, and cannot order the groups.
- **Example: Gender, place of residence, blood group, religion, division, ethnicity, marital status, hair color, etc.**

Ordinal Scale:

- The measurement scale, in which numbers are assigned to the categories or variable values for identification as well as ranking, is called an ordinal scale.
- In this scale, the numeric codes are used to identify the different categories of a variable in such a way that these numbers or values can be arranged in ascending or descending order.
- Example: **Socio-economic status** (poor, poorer, middle, richer, richest), **education level** (primary, secondary, higher – secondary), **income** (low, middle, high), **military rank**, etc.

Interval Scale:

- The measurement scale, in which numbers are assigned to the variable values in such a way that the level of measurement is broken down on a scale of equal units and the zero value on the scale is not absolutely zero (but a pre-set value) is called an interval scale.
- The interval scale measures its data along with the equal distance between two

adjacent values. This is the reason behind its name “interval”.

- **Example:** The variable temperature can have values 0° C, 10° C, 20° C etc. Here the value 0° C does not mean the absence of temperature. Thus, the value zero in interval scale is not absolutely zero.
- **Another example:** For an IQ test, one may use an interval scale. The use of an interval scale can help measure that the difference between an IQ of 80 and 90 is the same as an IQ of 90 and 100. At the same time, the score cannot be zero, because the minimum level for IQ is 40.

Ratio Scale:

- The measurement scale, in which numbers are assigned to the variable values in such a way that the level of measurement is broken down on a scale of equal units and the zero value on the scale is absolutely zero is called a ratio scale.
- In other words, we can say that when zero means absence of the characteristics then the appropriate scale to use is the ratio scale.
- Example: Age, weight, pulse rate etc.

Comparisons of the Measurement Scales:

Levels	Mathematical Operations	Features	Examples
Nominal	Counting	Categories	Gender
Ordinal	Counting Ranking	Categories Ranks	Socio-economic status
Interval	Counting Ranking Addition Subtraction	Categories Ranks Has equal units	IQ score
Ratio	Counting Ranking Addition Subtraction Multiplication Division	Categories Ranks Has equal units Has absolute zero	No. of children Age Weight

Exercise:

Specify the measurement scale of the following variables:

- a. Age (in years) of students in a class
- b. Nutritional status among children under 5 years of age
- c. Body mass index of adults
- d. Educational attainment of household heads
- e. Blood group of individuals

- f. Blood sugar level of diabetic patients
- g. Monthly income of people living in Dhaka
- h. Weights of newborn babies
- i. IQ score of children
- j. Ethnicity of people

Data Representation

Data: Information obtained by observing values of a variable is called data.

Raw Data:

- In any statistical investigation, when data first collected usually appear in raw form where, information has been recorded merely in arbitrary order.
- Data is called the raw data before the data are processed or ranked.
- Raw data, collected for any statistical investigation, is unable to represent the summary information.
- So it is necessary to represent the raw data in such a way, which will enable us to extract the preliminary ideas about the variable(s) under study.

Dealing with Raw Data:

The most frequently used methods for data representation are

- Tabulation
- Graphical representation

Tabulation:

A statistical method of data condensation by which we can represent summary information of one or more variables, is defined as tabulation.

Principles of the constructions of a table:

Some of the most basic principles that one should consider in constructing table are as follows:

1. The table should be self-explanatory. The title describing the contents of the table should be clear, concise and to the point.
2. The table should be as simple as possible. Two or three tables are often preferable to a large table containing too many details and variables.
3. The specified units of measurements for the data should be given.
4. Necessary code or symbols used in table should be explained in a footnote.
5. Sources of data should be mentioned.

1. Frequency Distribution Table:

- The number of times a particular value of a certain variable occurs in a set of observations is called the frequency of that value.
- And the manner in which the frequencies are distributed in the different classes is known as the frequency distribution of the values of that variable.

A sample of 100 students enrolled at a university were asked what they intended to do after graduation.

Table 2.3 Type of Employment Students
Intend to Engage In

Variable →	Type of Employment	Number of Students	← Frequency column
	Private companies/businesses	44	
Category →	Federal government	16	← Frequency
	State/local government	23	
	Own business	17	
		Sum = 100	

Example:

A sample of 30 employees from large companies was selected, and these employees were asked how stressful their jobs were. The responses of these employees are recorded below, where *very* represents very stressful, *somewhat* means somewhat stressful, and *none* stands for not stressful at all.

somewhat	none	somewhat	very	very	none
very	somewhat	somewhat	very	somewhat	somewhat
very	somewhat	none	very	none	somewhat
somewhat	very	somewhat	somewhat	very	none
somewhat	very	very	somewhat	none	somewhat

Construct a frequency distribution table for these data.

Table: Frequency distribution of 30 employee's job stress.

Stress on job	Frequency	Relative frequency	Percentage
Very	10	$10/30 = .333$	$.333 \times 100 = 33.3$
Somewhat	14	$14/30 = .467$	$.467 \times 100 = 46.7$
None	6	$6/30 = .200$	$.200 \times 100 = 20.0$
Total	30	1.00	100

The relative frequency

- The relative frequency of a category is obtained by dividing the frequency of that category by the sum of all frequencies.

$$\text{Relative frequency of a category} = \frac{\text{Frequency of that category}}{\text{Sum of all frequencies}}$$

- The relative frequency shows what fractional part or proportion of the total frequency belongs to the corresponding category.

The percentage

- The percentage for a category is obtained by multiplying the relative frequency of that category by 100.

$$\text{Percentage} = (\text{Relative frequency}) * 100$$

2. Contingency table:

- In statistics, a *contingency table* (also known as a cross tabulation or crosstab) is a type of *table* in a matrix format that displays the (multivariate) frequency distribution of the variables.
- A contingency table presents the results of two or more categorical variables. The joint responses are classified so that the categories of one variable are located in the rows and the categories of the other variable are located in the columns.
- The values located at the intersections of the rows and columns are called **cells**. Depending on the type of contingency table constructed, the cells for each row-column combination contain the frequency, the percentage of the overall total, the percentage of the row total, or the percentage of the column total.

Table: Frequency distribution of students by religion and gender

Religion	Gender		Total
	Male	Female	
Muslim	25	20	45
Hindu	12	12	24
Christian	8	6	14
Buddha	5	3	8
Others	2	2	4
Total	52	43	95

Test Yourself: A sample of 500 shoppers was selected in a large metropolitan area to determine various information concerning consumer behavior. Among the questions asked was “do you enjoy shopping for clothing?” the results are summarized in the following cross classified table:

Table: Frequency distribution of preference of shopping for clothing of the consumer

Enjoy shopping for clothing	Gender		Total
	Male	Female	
Yes	136	224	360
No	104	36	140
Total	240	260	500

- a) Construct a contingency table based on total percentages.
- b) Determine the proportion
 - i. Male among total respondents
 - ii. Female among total respondents
 - iii. Proportion of people who enjoys shopping
 - iv. Proportion of people who does not enjoys shopping
- c) Determine the proportion (or percentage) of
 - i. Male among those who enjoy shopping.
 - ii. People who enjoys shopping among male

Graphical Representation of Data

Importance:

- The graph is easiest to understand as the information portrayed is in facts and figures.
- It allows us to relate and compare data for different time periods.
- It is used in statistics to determine the mean, median and mode of different data.
- It saves a lot of time as it covers most of the information in facts and figures.

Different types of charts and graphs:

Data charts are available in a wide variety of maps, diagrams, and graphs. Selection of the most appropriate chart depends on a variety of different factors, such as –

- i. Objective of the researcher
- ii. The nature of the data,
- iii. The purpose of the chart, and
- iv. Types of variable etc.

For example, some of the most frequently graphs are used for

Qualitative variable	• Pie Chart • Bar Diagram
Quantitative variable	• Histogram • Frequency polygon • Percentage polygon • Cumulative frequency polygon • Line plot • Dot plot • Ogive • Stem and leaf plot

Organizing and Graphing Qualitative Data

Before constructing graph, data should be organized using frequency distribution table.

Table 2.5 Relative Frequency and Percentage Distributions of Stress on Job

Stress on Job	Relative Frequency	Percentage
Very	$10/30 = .333$	$.333(100) = 33.3$
Somewhat	$14/30 = .467$	$.467(100) = 46.7$
None	$6/30 = .200$	$.200(100) = 20.0$
	Sum = 1.000	Sum = 100

Pie Chart

- A pie chart is a disk divided into wedge-shaped pieces proportional to the relative frequencies of the qualitative data.
- Pie Diagram will give appropriate representation of the data when the variable of interest is categorical (qualitative or discrete quantitative).

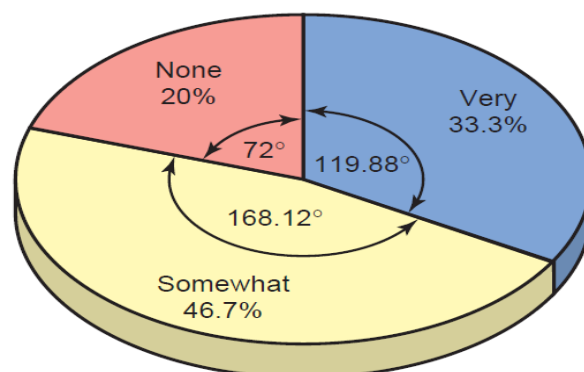


Table 2.6 Calculating Angle Sizes for the Pie Chart

Stress on Job	Relative Frequency	Angle Size
Very	.333	$360(.333) = 119.88$
Somewhat	.467	$360(.467) = 168.12$
None	.200	$360(.200) = 72.00$
Sum = 1.000		Sum = 360

To construct a Pie chart:

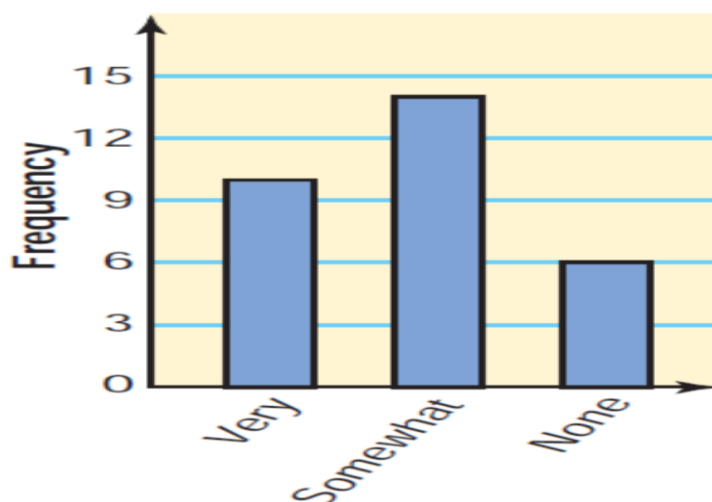
- Obtain a relative frequency distribution of the data.
- Divide a disk into wedge-shaped pieces proportional to the relative frequencies.
- Label the slices with the distinct values and their relative frequencies.

Some drawbacks of Pie chart:

- If there are too many categories then Pie Diagram is not a good option.
- It is difficult to compare two or more data sets.
- Pie chart is considered bad for visualization as humans can compare length better than angles.

Bar Diagram

- Frequencies, relative frequencies, or percentages can be used to label a bar chart. Although we primarily use relative frequencies, some of our applications employ frequencies or percentages.
- A bar chart displays the distinct values of the qualitative data on a horizontal axis and the relative frequencies (or frequencies or percentages) of those values on a vertical axis.
- The relative frequency of each distinct value is represented by a vertical bar whose height is equal to the relative frequency of that value.
- **The bars should be positioned so that they do not touch each other.**

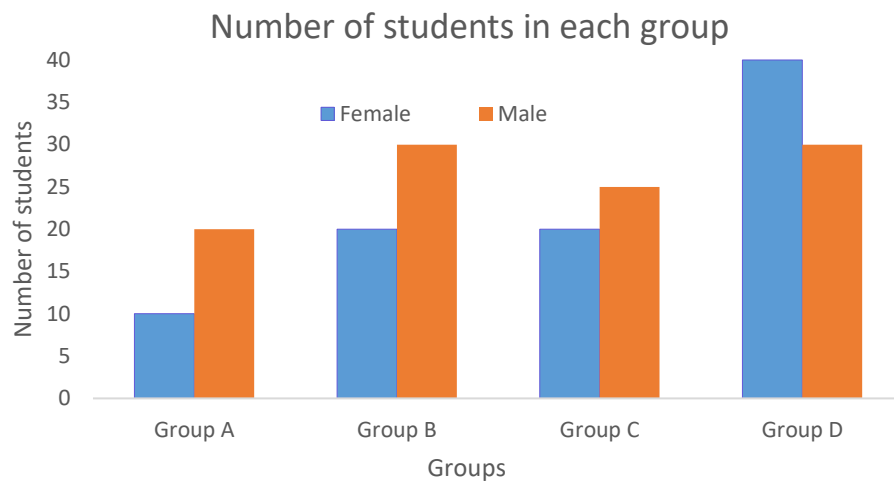


To construct a Bar chart:

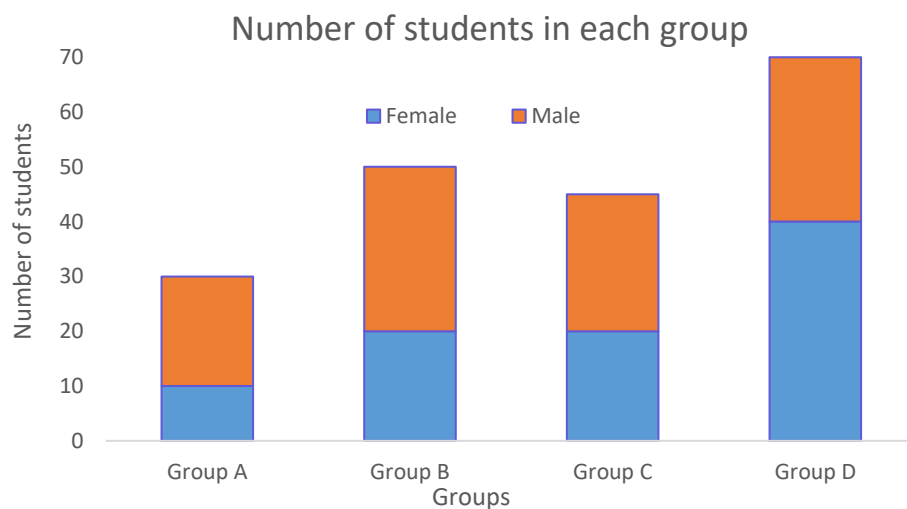
1. Obtain the frequency distribution of the data
2. Draw a horizontal axis on which to place the bars and a vertical axis on which to display the frequencies.
3. For each distinct value of variable, construct a vertical bar whose height equals the frequency of that value.
4. Label the bars with the distinct values, the horizontal axis with the name of the variable, and the vertical axis with frequency.

Bar plot can easily be used for comparing two or more data sets.

1. Clustered/ Grouped bar:



2. Stacked bar:



Organizing and Graphing Quantitative Data

Frequency distribution:

- A frequency distribution for quantitative data lists all the classes and the number of values that belong to each class.
- And the manner in which the frequencies are distributed in the different classes is known as the frequency distribution of the values of that variable.

Table 01: Frequency distribution of number of children per family

(ungrouped data)

Number of children	Number of families
0	10
1	27
2	15
3	18
4	9

Table 02: Frequency distribution of height of trees in Sundarbans

(grouped data)

Height of the tree (In Feet)	Number of trees
0-50	1000
50 – 100	2735
100 – 150	1589
150 – 200	1518
More than 200	719

For grouped data:

1. Class limit: Class limits are the highest and the lowest values that can be included in the class.

For example, if we consider the class 50 – 100, here 50 is the lower limit and 100 is the upper limit. In such case no values greater than 100 shall fall into that class. Similarly, no values less than 50 shall fall into that class either.

2. Class interval: The difference between the upper limit and the lower limit of a class is called the class interval. Class interval is usually denoted by c , i , h or w .

For example, the class interval of the class ‘50 – 100’ is 50.

3. Class frequency: The number of observations falling within a particular class is called its frequency or class frequency.

4. Class midpoint or class mark: The value of the variable that lies in the middle of the upper and lower limits is called mid value or midpoint of the class.

It can be obtained as follows:

Class midpoint = $\frac{l+s}{2}$; Where l = Lower limit of the class, s = Upper limit of the class

5. Relative frequency (also known as proportion): Instead of presenting the frequencies in absolute terms, it is sometimes convenient to express the frequencies in percentages. The relative frequency (also known as proportion) corresponding to a class is simply the ratio of the total number of items in that class to the total number of elements in the total set. **Multiplying relative frequency by 100 one can obtain the percentage of observation that belongs to any particular class.**

Relative frequency = Proportion = Frequency in each class / Total number of frequency

6. Cumulative frequency: The cumulative frequency corresponding to a class is the total of all frequency up to and including that class.

Exercise: Let us consider the following table showing the distribution of marks of 27 students

Class limit	Class mid value	Frequency	Relative frequency	Cumulative frequency	Cumulative relative frequency
0 – 10	5	4	0.148	4	0.148
10 - 20	15	8	0.296	4+8	0.444
20 – 30		5		4+8+5	
30 – 40		4			
40 – 50		3			
50 – 60		2			
60 - 70	65	1			
Total		27			

Example:

Marks obtained by 20 students in the class test are as follows: 5, 10, 20, 15, 5, 20, 20, 15, 15, 15, 10, 10, 10, 20, 15, 5, 18, 18, and 18.

Table: Frequency distribution table for ungrouped data.

Marks obtained in Test	No. of Students
5	3
10	4
15	5
18	4
20	4
Total	20

Table: Frequency distribution table for grouped data.

Marks obtained in Test	No. of Students
1 - 5	3
6 - 10	4
11 - 15	5
16 - 20	8
Total	20

General Rules for forming Frequency Distribution

1. Determine the largest and smallest numbers in the raw data and thus find the range (difference between largest and smallest numbers).
2. Determine the number of classes.
3. Calculate class interval using the following formula:
$$\text{Class interval} = i = (\text{Highest Value} - \text{Lowest Value}) / \text{Number of classes}$$
4. Determine the number of observations falling into each class interval, i.e. find the class frequencies.

Determine the Number of classes

- One's professional judgment can determine the number of classes.
- Too many classes or too few classes might not reveal the basic shape of the set of data.
- As a general rule, it is best to not use less than 5 nor more than 15 classes in the construction of a frequency distribution.
- The number of classes can be estimated based on the number of observations (n), using one of the rules:
 1. **The 2^k rule:** As the number of classes, select the smallest integer, k (whole number) such that $2^k \geq n$.
 2. **Alternatively,** estimate the number of classes using formula:
$$\text{Number of classes} = 1 + 3.322 \times \log_{10}(n)$$

Example: The following data represent the number of t-shirts sold over last 20 weeks

1 5 6 7 8 8 9 10 11 11 12 12 14 14 15 15 17 17 18 19

Construct a frequency distribution table with appropriate class interval.

Step 1: Highest value = 19, Lowest value = 1

$$\text{So, range} = 19 - 1 = 18$$

Step 2: Here, n = 20

$$\text{Using } 2^k \text{ rule, } 2^5 = 32 > 20$$

$$\text{So, no. of classes} = 5$$

$$\text{Step 3: class interval} = (19 - 1) / 5 = 18/5 = 3.6 = 4$$

Step 4:

Table: Frequency Distribution of Number of T-shirt sold

No. of t-shirts sold	Frequency	Relative Frequency
1 – 4	1	1/20
5 – 8	5	5/20
9 – 12	6	6/20
13 – 16	4	4/20
17 – 20	4	4/20
Total	20	1.00

Test Yourself: The profit in (lakh of taka) of 50 companies for the year 2017-18 are given below:

66	62	45	79	32	51	56	60	51	49
25	42	54	54	58	70	43	58	52	50
38	67	50	59	48	65	71	30	46	55
82	51	63	45	53	40	35	56	70	52
67	55	57	30	63	42	74	58	44	55

Construct a complete frequency distribution table for the above data.

Test Yourself: The following data specifies the life of 40 similar car batteries recorded to the nearest tenth of a year:

2.2, 4.1, 3.5, 4.5, 3.2, 3.7, 3.0, 2.6, 3.4, 1.6, 3.1, 3.3, 3.8, 3.1, 4.7, 3.7, 2.5, 4.3, 3.4, 3.6, 2.9, 3.3, 3.9, 3.1, 3.3, 3.1, 3.7, 4.4, 3.2, 4.1, 1.9, 3.4, 4.7, 3.8, 3.2, 2.6, 3.9, 3.0, 4.2, 3.5

Construct a complete frequency distribution table.

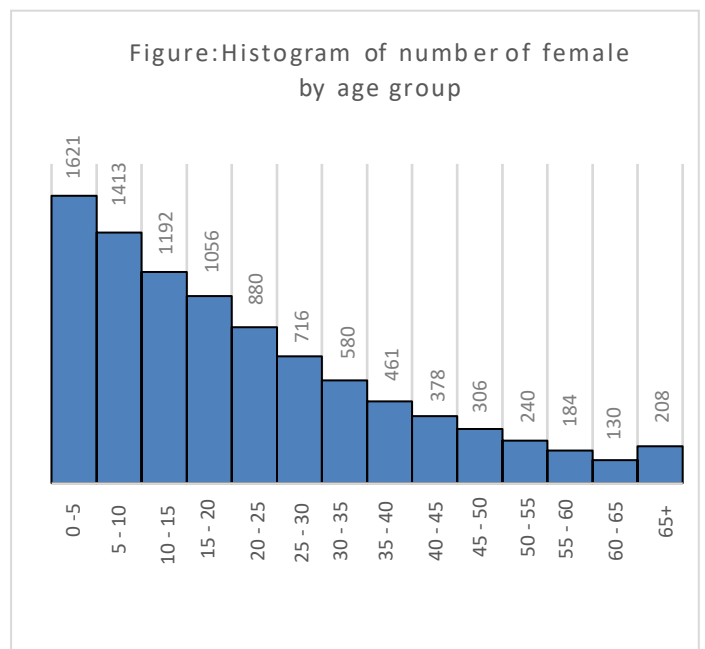
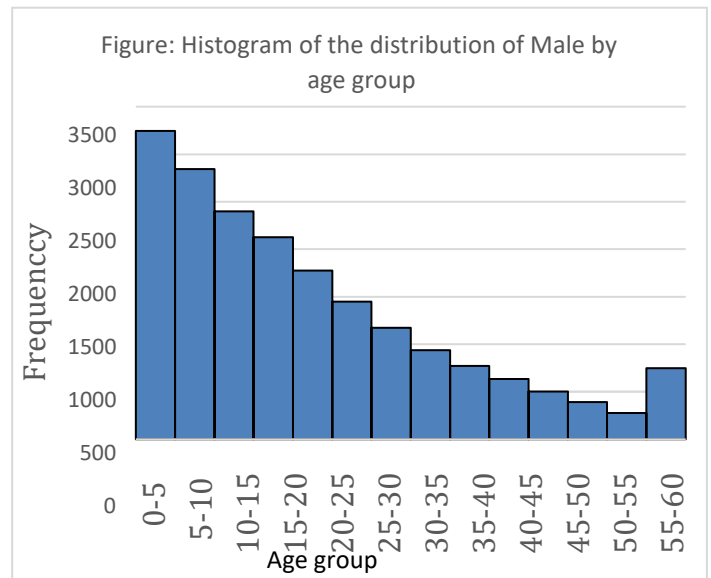
Histogram

- A **histogram** is a bar chart for **grouped numerical** data in which the frequencies (relative frequencies, percentages) of each group of numerical data are presented as individual vertical bars.
- The variable of interest is displayed or plotted along the horizontal (X) axis.
- Frequency (relative frequencies, percentages) of the values per class is displayed or plotted along the vertical (Y) axis.
- **The bars should be positioned so that they touch each other.**

Note: In a histogram there are no gaps between adjacent bars but in a bar chart of categorical data there is a gap between adjacent bars.

Example:

Table : frequency distribution of male and female by age group		
Age group	Male	Female
0-5	3243	1621
5-10	2842	1413
10-15	2398	1192
15-20	2125	1056
20-25	1776	880
25-30	1450	716
30-35	1173	580
35-40	936	461
40-45	773	378
45-50	633	306
50-55	503	240
55-60	391	184
60-65	278	130
65+	749	208



To Construct a Histogram:

1. Obtain a frequency (relative-frequency, percent) distribution of the data.
2. Draw a horizontal axis on which to place the bars and a vertical axis on which to display the frequencies (relative frequencies, percentages).
3. For each class, construct a vertical bar whose height equals the frequency (relative frequency, percent) of that class.
4. Label the bars with the classes, the horizontal axis with the name of the variable, and the vertical axis with Frequency (Relative frequency, Percent).

Note: Relative-frequency (or percent) histograms are better than frequency histograms for comparing two data sets because the same vertical scale is used for all relative frequency histograms (a minimum of 0 and a maximum of 1) making direct comparison easy.

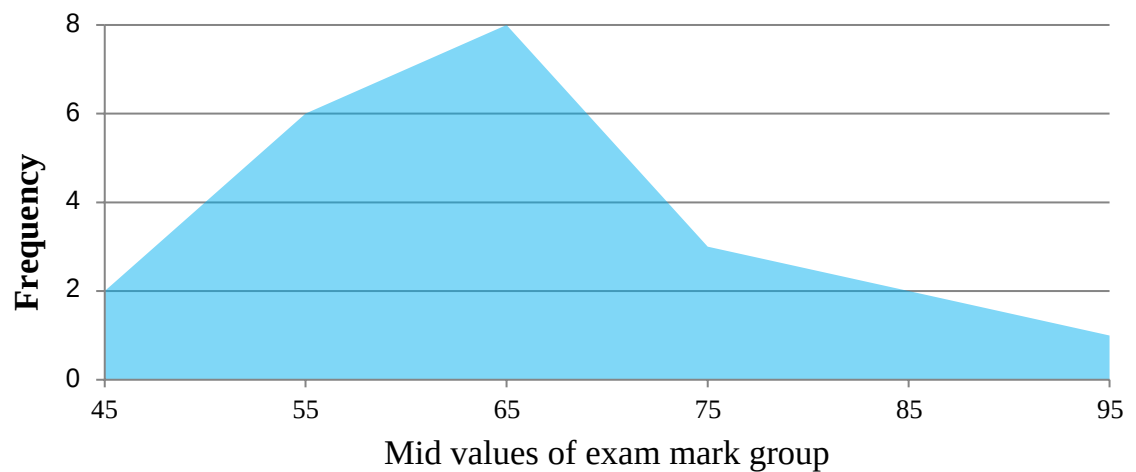
The Frequency Polygon

In constructing frequency polygon –

- The mid values of the class intervals of the frequency distribution are placed on the horizontal (X) axis
- And the corresponding frequencies are represented on the vertical (Y) axis.
- The co-ordinates points thus obtained joined by straight line.

Table 1.4: Frequency distribution of marks		
Marks	Mid value	Frequency
40-50	45	2
50-60	55	6
60-70	65	8
70-80	75	3
80-90	85	2
90- 100	95	1

Figure: Frequency distribution of marks obtained by students



The Percentage Polygon

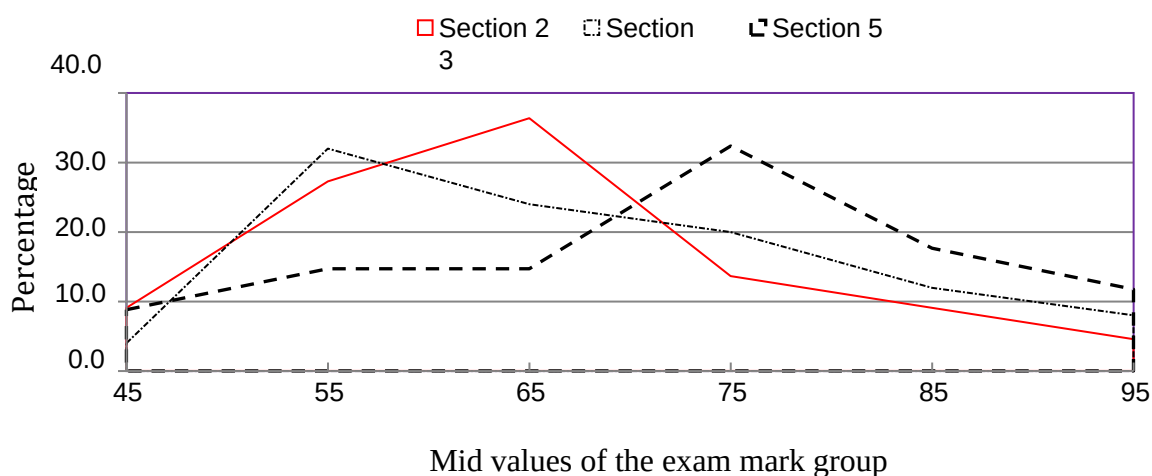
When there are two or more groups, one should use a percentage polygon. Because constructing multiple histograms on the same graph to compare two or more data sets often gets confusing. Super imposing the vertical bars of one histogram on another histogram makes interpretation difficult.

- A percentage polygon is formed by having the midpoint of each class represent the data in that class
- Then connecting the sequence of midpoint at their respective class percentages.

The following table and figure illustrates the construction of the percentage polygon.

Table: Frequency distribution of Marks obtained by students taught by “X”							
Mark Group	Mid value	Frequency of students			Percentage of Students		
		Section 2	Section 3	Section 5	Section 2	Section 3	Section 5
40-50	45	2	1	3	9.1	4.0	8.8
50-60	55	6	8	5	27.3	32.0	14.7
60-70	65	8	6	5	36.4	24.0	14.7
70-80	75	3	5	11	13.6	20.0	32.4
80-90	85	2	3	6	9.1	12.0	17.6
90-100	95	1	2	4	4.5	8.0	11.8
Total		22	25	34	100	100	100

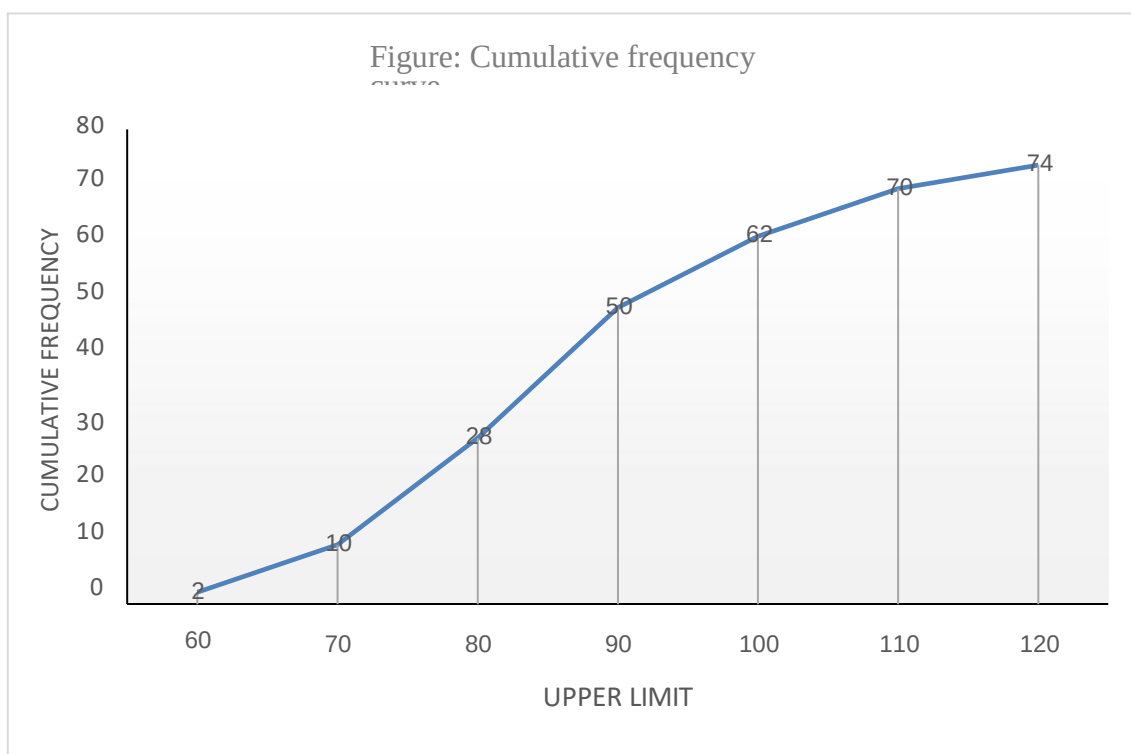
Figure: Comparison of percentage distribution of grades obtained by students



Ogive or Cumulative frequency curve:

- Ogive or a cumulative frequency graphs is drawn by plotting the cumulative frequency (or relative cumulative frequency) of a class against the upper limit of the corresponding class.
- In these graphs the exact limits of the class intervals are plotted along the X-axis and the cumulative frequencies (or relative cumulative frequencies) are plotted along the Y-axis.
- Coordinate points thus obtained is then joined with a line drawing smoothly. This will result in curve called ogive or cumulative frequency curve.
- **Such curve is usually used to determine the values of certain quantities such as median, quartile, percentile etc.**

Class interval of the amount of sell	upper limit	frequency	Cumulative frequency	Cumulative relative frequency
50 - 60	60	2	2	$2/74 = 0.027$
60 - 70	70	8	10	$10/74 = 0.135$
70 - 80	80	18	28	$28/74 = 0.378$
80 - 90	90	22	50	0.676
90 - 100	100	12	62	0.838
100 - 110	110	8	70	0.946
110 - 120	120	4	74	1.00



The Stem and Leaf Display

- Statisticians continue to invent ways to display data.
- One method, developed in the 1960s by the late Professor John Tukey of Princeton University, is called a stem and leaf diagram.
- This diagram is often easier to construct than either a frequency distribution or a histogram and generally displays more information.
- In a stem and leaf diagram each observation is separated into two parts, namely, a stem consisting of all but the rightmost digit and a leaf, the rightmost digit.

To Construct a Stem-and-Leaf Diagram

- Each numerical value is divided into two parts, all but the rightmost digit(s) becomes the stem and the rightmost digit the leaf.
- Write the stems from smallest to largest in a vertical column to the left of a vertical rule.
- Write each leaf to the right of the vertical rule in the row that contains the appropriate stem.
- Arrange the leaves in each row in ascending order.

Example:

The following data represented the marks obtained by 20 students in a statistics test.

84	17	78	45	47	53	76	54	75	22
66	65	55	54	51	33	39	19	54	72

Use the stem and leaf plot to display the data

The stem leaf plot for the given		After arranging the stem leaf plot we get for the given data	
Stem	Leaf	Stem	Leaf
1	7,9	1	7,9
2	2	2	2
3	3,9	3	3,9
4	5,7	4	5,7
5	3,4,5,4,1,4	5	1,3,4,4,4,5
6	6,5	6	5,6
7	8,6,5,2	7	2,5,6,8
8	4	8	4

The stem and leaf plot technique (a graphical technique of representing quantitative data) allows us to use the information contained in a frequency distribution to show

- The range of the values
- Point of concentration of scores
- The shape of the frequency distribution
- Whether there are any extreme values in the distribution.

Exercise: Cholesterol levels for 20 high-level patients are given below:

210	209	212	208
217	207	210	203
208	210	210	199
215	221	213	218
202	218	200	214

Construct a stem and leaf diagram for these data.

Exercise: Construct a stem and leaf plot for these data -

74 89 54 80 93 64 77 72 70 66 85 89 81 56 81 71 74 82 85 63 72 81 81
95 84 81 52 80 70 69 66 60 83 85 98 84 68 90 82 79 72 87 88

Reference books

1. Introductory Statistics by Neil A. Weiss
2. Introductory Statistics by Prem S. Mann

Thank you